



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2023-24(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

End Semester

Semester: V

Programme: B. Tech Information Technology

Course: Theory of Computation

Time: 3 Hour

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33502	Course Outcome
CO1	To illustrate finite state machines to solve problems in computing.
CO2	To classify theoretical foundations of computer science from the perspective of formal languages.
CO3	To familiarize Regular grammars and Context Free Grammars.
CO4	To enhance students' ability to understand and conduct mathematical proofs for computation and algorithms.
CO5	To explain hierarchy of problems arising in computer science.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define NFA.	CO1	2	2
b.	What is pumping lemma?	CO2	2	2
c.	Define PDA.	CO3	2	2
d.	What is TM?	CO4	3	2
e.	Define decidability.	CO5	2	2
Group B				

Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain DFA and NFA with differences.	CO1	2	5
b.	Convert NFA to DFA.	CO1	2	5
c.	Explain minimization of FA.	CO1	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain pumping lemma with example.	CO2	2	5
b.	Convert RE to DFA.	CO2	3	5
c.	Explain closure properties.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain CFG and derivation tree.	CO3	2	5
b.	Convert CFG to CNF/GNF.	CO3	3	5
c.	Explain PDA and its working.	CO3	3	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain TM model.	CO4	2	5
b.	Explain recursive vs recursively enumerable languages.	CO4	3	5
c.	Explain LBA.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Halting problem.	CO5	2	5
b.	Explain PCP.	CO5	3	5
c.	Explain recursive functions.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create